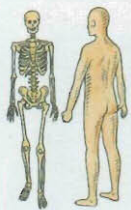
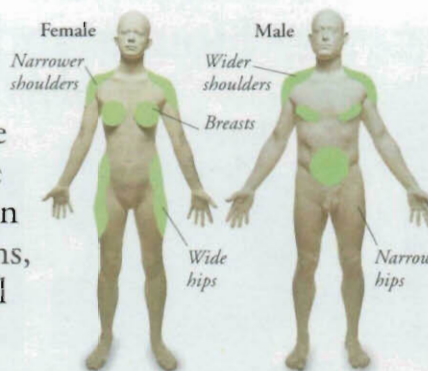


HUMAN BODY



ALTHOUGH WE ALL LOOK different, we are identical in the way our bodies are constructed and function. Each human body is built up from 12 major systems, including the digestive system, skeletal system, and muscular system. These systems interact to produce co-ordinated, active, intelligent humans. The study of the body's structure is called anatomy. Externally, the only consistent anatomical differences between humans are between males and females.

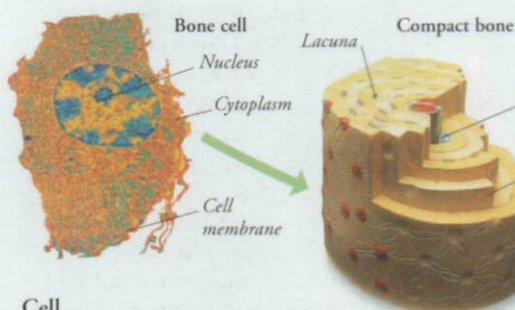


Anatomy

The human body is divided into the head and neck, the trunk (consisting of the chest, abdomen, and pelvic region), and the arms and the legs. Men and women differ in their external genitals and in the places where fat accumulates (shown in green).

From cells to systems

The body's billions of cells are organized into tissues. Each tissue consists of similar types of cell. One or more types of tissue work together inside an organ, such as a bone or a lung. Organs are linked together to form a system that has one or more major roles. Together the systems are collected together to form the body.



Cell

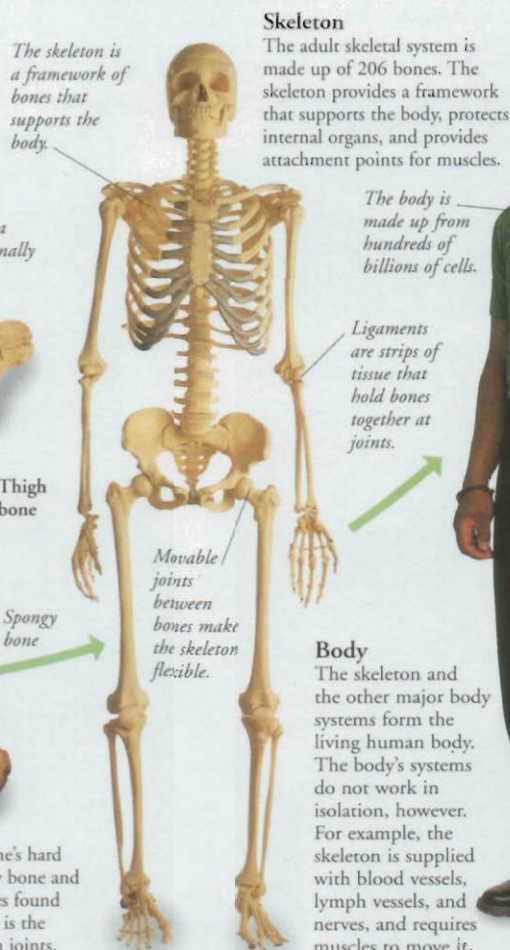
Osteocytes, or bone cells, are spider-shaped cells that make up the tissues that form a bone. Osteocytes are found in spaces called lacunae that are scattered about the hard matrix (material) found in bone tissue. Their job is to maintain the hard matrix.

Tissue

Compact bone is one of the tissues that makes up a bone. It consists of layers of hard bone around a central tube called the Haversian canal. This carries blood vessels which supply the osteocytes with food and oxygen.

Organ

Each bone consists of different tissues. Compact bone is a bone's hard outer covering. Spongy bone and bone marrow are tissues found inside bones. Cartilage is the slippery tissue found in joints.



Body

The skeleton and the other major body systems form the living human body. The body's systems do not work in isolation, however. For example, the skeleton is supplied with blood vessels, lymph vessels, and nerves, and requires muscles to move it.

Body systems

Each body system contributes to the body's normal functioning. Together, the body's systems are controlled by the nervous and endocrine systems. They enable us to move, talk, and perceive the world, while our internal processes run automatically.



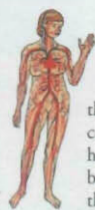
Muscular

The muscular system moves and supports the body. It consists of over 620 skeletal muscles attached to bones.



Nervous

The nervous system controls the body's activities. It consists of the brain and spinal cord, and a network of nerves.



Circulatory

The circulatory system transports material around the body. It consists of the heart, a network of blood vessels, and the blood.



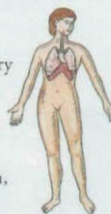
Digestive

The digestive system supplies the body with food. It consists of the mouth, oesophagus, stomach, and intestines.



Integumentary

The integumentary system is the body's outer, protective covering and consists of skin, hair, and nails.



Respiratory

The respiratory system supplies the body with oxygen. It consists of the nose, throat, trachea, and lungs.



Urinary

The urinary system removes waste. It consists of the two kidneys, the ureters, the bladder, and the urethra.



Endocrine

The endocrine system regulates many body processes. It consists of glands that make hormones.



Lymphatic

The lymphatic system protects the body against disease. It consists of a network of lymph vessels.



Reproductive

The reproductive system enables us to produce children. Male and female systems are different.

FIND OUT
MORE

BRAIN AND
NERVOUS SYSTEM

CELLS

GROWTH AND
DEVELOPMENT

HEART AND
CIRCULATORY SYSTEM

HUMAN
EVOLUTION

MUSCLES AND
MOVEMENT

REPRODUCTION

SKELETON

SKIN, HAIR,
AND NAILS